

Professional Journey Overview

Software Engineer 1 (Full-Time) | 2 Internships (Summer + Winter) → FTE Offer

Delivered **\$27M+ annual value** across three major initiatives spanning internships and full-time employment. Specialized in AI/ML-powered enterprise platforms, distributed systems, and full-stack development with measurable impact in cost reduction, predictive analytics, and operational efficiency.

Summer SDE Internship: User360 Web Application | TMX Data Platform

Project Overview

End-to-end development of **User360 Web Application** for Dell's Modern Application and Data Team, providing comprehensive employee data access and management across the TMX Data Platform.

Technology Stack

Backend: Python, FastAPI • **Frontend:** React.js, JavaScript • **Database:** Greenplum, PostgreSQL • **DevOps:** GitLab CI/CD

Core Technical Contributions

1. Full-Stack Application Development

- Built interactive dashboard with intuitive UI design for comprehensive employee details
- Enhanced data accessibility and information retrieval across the organization
- Implemented responsive components for seamless user experience

2. Database Integration & API Development

FastAPI Backend:

- Developed RESTful APIs for data retrieval from Greenplum & PostgreSQL
- Optimized query performance for substantial datasets
- Implemented connection pooling for concurrent requests
- Reduced data retrieval time through efficient caching strategies

API Performance Improvements:

Before: Sequential queries → 3.2 sec average response time
After: Optimized queries + caching → 0.4 sec response time (-88%)
Achieved through: Query optimization (EXPLAIN ANALYZE),
indexed columns (B-tree on user_id, timestamp),
Redis caching (TTL=300s, hit rate 76%),
connection pooling (min=5, max=20 connections)

Data Retrieval Efficiency:

Before: Full table scans → 45 sec for 500K records
After: Partitioned queries + pagination → 1.8 sec (-96%)
Achieved through: Table partitioning (monthly),
cursor-based pagination (limit=1000),
parallel query execution,
materialized views for aggregations

Throughput:

Before: 50 requests/sec → bottleneck at database layer
After: 340 requests/sec (+580%) → linear scaling to 500 req/sec

3. CI/CD Pipeline Optimization

GitLab CI/CD:

- Streamlined deployment workflows
- Automated testing and build processes
- 40% reduction in deployment time
- Enabled seamless updates with zero-downtime releases

Outcome: Awarded **6-month Winter Internship + FTE Offer** from Dell Technologies upon completion

Winter SDE Internship: Security Governance & Training Portal

Project Scope

Led frontend development for **Dell Security Governance Web App** and contributed to **Digital People Training Portal** with AI-enabled features for Team Member Experience team.

Technology Stack

Frontend: React.js, ATOM architecture • **Backend:** Node.js, GraphQL, FastAPI • **AI/ML:** Elasticsearch, LangChain, MLflow • **Infrastructure:** Docker, Kubernetes (AKS), Redis, PostgreSQL • **DevOps:** Azure DevOps, Terraform

Core Technical Contributions

1. Security Governance Web App (Frontend from Scratch)

ATOM Architecture Implementation:

- Atomic design pattern (Atoms, Molecules, Organisms, Templates)
- Clean, reusable component library
- Dynamic content fetching from API layer
- Modular codebase for scalability and maintainability

2. Digital People Training Portal Development

Database Architecture:

- Collaborated on schema design and table creation
- Established relational data models and foreign key constraints
- Optimized indexing for query performance

GraphQL API Development:

```
type Query {  
  trainingModules(filter: TrainingFilter): [TrainingModule]  
  userProgress(userId: ID!): UserProgress  
}
```

Benefits:

- Optimized data querying with precise field selection
- Reduced over-fetching and under-fetching
- Enhanced system efficiency for seamless user experiences

3. AI-Enabled Features Development

Semantic Search Engine:

Technology Stack: Elasticsearch + LangChain + MLflow

- Vector embeddings for semantic understanding
- Natural language query processing
- Relevance scoring with ML-based ranking
- Integration with internal tooling ecosystem

Search Accuracy Metrics:

Before: Keyword matching → 58% user satisfaction, avg 8 queries/search

After: Semantic embeddings → 84% satisfaction (+26 pts), avg 2.3 queries/search

Achieved through: Sentence-transformers (all-MiniLM-L6-v2),
fine-tuning on 12K training queries,
negative sampling, BM25 + vector hybrid retrieval ($\alpha=0.7$)

Precision@10: 0.67 → 0.89 (+33% improvement)

Mean Reciprocal Rank (MRR): 0.54 → 0.82 (+52% improvement)

Recommendation System:

Collaborative filtering for personalized training suggestions

Content-based recommendations using feature engineering

Real-time inference with Redis caching

MLflow for model versioning and A/B testing

Recommendation Accuracy:

Before: Rule-based suggestions → 23% click-through rate

After: Hybrid CF + content model → 41% CTR (+78% relative improvement)

Achieved through: Matrix factorization (ALS, rank=50),
user-item interaction matrix (sparse, 94% sparsity),
content features (TF-IDF + metadata),
weighted hybrid ($\lambda=0.6$ collaborative, 0.4 content)

NDCG@5: 0.48 → 0.74 (+54% improvement)

Coverage: 45% → 78% (long-tail item exposure)

4. CI/CD Pipeline Transformation

Azure DevOps Pipeline Revamp:

Before: Multiple job failures, low reliability

After: 90%+ maturity score, enhanced performance

Improvements:

- Fixed failing stages across build, test, and deploy
- Implemented comprehensive error handling
- Added automated rollback mechanisms
- Integrated quality gates and security scanning

Pipeline Reliability Metrics:

Before: 58% success rate, avg 45 min build time, manual interventions

After: 94% success rate (+36 pts), avg 12 min build (-73%), zero manual fixes

Achieved through: Parallel job execution (3 agents),
Docker layer caching (BuildKit),
incremental builds (artifact reuse),
flaky test isolation and retry logic (max 3 retries)

Quality Gates Impact:

Before: 32% code coverage, no security scanning

After: 82% code coverage (+50 pts), 100% vulnerability scanning

Achieved through: Jest/Pytest integration in pipeline,
SonarQube quality gates (blocking on <80% coverage),
OWASP dependency check, Trivy container scanning

Deployment Frequency:

Before: Weekly releases → 8-12 hour deployment window

After: Daily releases → 15-minute deployment window (-98%)

Mean Time to Recovery (MTTR): 4 hours → 22 minutes

Impact: Significant enhancement in deployment reliability and team productivity

Full-Time: Dell Communique | \$25M Enterprise Communication Platform

Architecture & Technology Stack

Frontend: React • **Backend:** Node.js, Go • **Infrastructure:** RabbitMQ, Docker, PKS/Kubernetes, SMTP • **AI/ML:** Azure OpenAI (GPT-4/5), Llama 3.3, XGBoost, TensorFlow, PyTorch • **Data:** Azure Synapse, Data Lake Gen2, Delta Tables, Redis • **Analytics:** Power BI, MLflow

Core Technical Contributions

1. AI-Powered Personalization Engine

Data Pipeline Architecture:

Azure Data Factory → Delta Lake → Feature Engineering (Databricks/PySpark)

↓

[ML Model Ensemble]

- Segmentation: K-Means, DBSCAN, Gaussian Mixture
- Interest Profiling: Sentence-BERT embeddings, TF-IDF vectors
- Recommendation: Matrix Factorization + BERT (collaborative+content-based)

- └ Behavioral Prediction: XGBoost, Random Forest (open/click probability)
- └ GenAI Optimizer: Azure OpenAI GPT-4/5 (tone & variant rewriting)



FastAPI/Flask → Redis Cache → Communique Rendering (per-user variants)

Key ML Implementations:

- **Micro-segmentation:** Unsupervised clustering discovering hidden groups ("tech-savvy APJ new joiners")
- **Multi-armed bandit:** Dynamic variant selection learning best content per audience
- **Reinforcement learning:** Adaptive optimization from continuous engagement feedback
- **Feature Store:** Azure ML Feature Store with nightly retraining on engagement data

ML Performance Metrics:

Engagement Prediction Accuracy:

Before: Manual segmentation → 52% open rate prediction accuracy

After: XGBoost ensemble model → 78% accuracy (+26 percentage points)

Achieved through: Historical engagement features (30+ dimensions),
interaction timestamps, behavioral patterns, A/B test data

Personalization Impact:

Before: Generic broadcasts → 18% average open rate, 2.3% click rate

After: ML-driven personalization → 25% open rate (+39%), 4.1% click rate (+78%)

Achieved through: Sentence-BERT embeddings, real-time user profiling,
multi-armed bandit optimization (epsilon-greedy: $\epsilon=0.1$)

Topic Classification Accuracy:

Before: Manual tagging → 65% topic relevance accuracy

After: BERTopic + LDA hybrid → 89% accuracy (+24 percentage points)

Achieved through: Pre-trained transformers, hierarchical clustering,
domain-specific fine-tuning on 50K+ internal communications

Impact: 40% engagement lift • 60% faster template creation • Hyper-personalized delivery at 100K+ scale

2. Predictive Analytics & Intelligence Layer

[Predictive Models]

- └ Engagement Prediction (XGBoost/Gradient Boosted Trees)
 - └ → Pre-send open/click probability forecasting
- └ Attrition Correlation (Cox Proportional Hazard, Survival Analysis)
 - └ → Links low engagement → burnout/attrition risk scores

- |— Topic Effectiveness (LDA, BERTopic, spaCy)
 - | → Multi-dimensional analysis (region, role, channel)
- |— ROI Measurement (Propensity Score Matching, Causal Impact)
 - | → Quantified campaign impact on adoption/retention
- |— Anomaly Detection (Isolation Forest, Z-score)
 - | → Real-time dips/spikes alerting via Azure Monitor

AI KPI Assistant:

- LLM-based summarization generating natural language insights
- Auto-generated executive reports ("Engagement up 12% MoM, driven by APJ product updates")
- Template-driven prompt engineering for consistent narratives

Predictive Analytics Performance:

Attrition Correlation Model:

Before: No predictive capability → Reactive HR interventions

After: Cox regression model → 0.82 AUC-ROC, 74% precision @ 80% recall

Achieved through: Survival analysis on 2-year historical data,
engagement velocity features, interaction depth metrics,
integration with Workday attrition events (N=3,400 records)

ROI Measurement Accuracy:

Before: Attribution guesswork → Uncertain campaign impact

After: Propensity Score Matching → 91% confidence intervals on lift measurement

Achieved through: Matched cohorts (treatment vs. control),
causal inference techniques, confounding variable adjustment,
validation against A/B test ground truth (15 campaigns)

Anomaly Detection Precision:

Before: Manual monitoring → 40% false positive rate, delayed detection

After: Isolation Forest → 88% precision, 92% recall, <5 min detection latency

Achieved through: Time-series decomposition, rolling window statistics,
adaptive thresholds (Z-score > 3.5), Azure Monitor integration

3. Go Microservices Architecture

// High-performance backend services

- SharePoint API integration (migrated from S3)
- Binary signature validation + concurrent uploads (goroutines)
- RabbitMQ queue management:
 - * Standard Queue: Batch processing (10K emails/min)

- * Priority Queue: Time-sensitive communications
- * Dead Letter Queue: Retry logic (3 attempts, exponential backoff)
- SMTP connection pooling for guaranteed delivery (99.9% rate)

Go Service Performance:

Before: Node.js S3 uploads → 850ms p95 latency, 45 req/sec throughput

After: Go SharePoint service → 180ms p95 latency (-79%), 320 req/sec (+611%)

Achieved through: Goroutine pooling (worker pool: 50 concurrent),

HTTP/2 multiplexing for SharePoint,
zero-copy buffer management,
memory pooling (sync.Pool for byte buffers)

Upload Reliability:

Before: 94.2% success rate, no retry mechanism

After: 99.9% success rate (+5.7 pts), intelligent retry with backoff

Achieved through: Exponential backoff (initial: 1s, max: 32s),

circuit breaker pattern (failure threshold: 5),
context-aware timeouts (30s per request),
health check endpoints for monitoring

Email Delivery Metrics:

Before: 96.8% delivery, 4.2% bounces/failures

After: 99.9% delivery (+3.1 pts), 0.1% failure rate

Queue Processing: 10K emails/min sustained, 15K burst capacity

Achieved through: Connection pooling (20 SMTP connections),

batch sending (100 emails/batch),
adaptive rate limiting based on server response

4. Full-Stack Analytics Dashboard (React + Node.js)

- Real-time KPI tracking: open rates, click heatmaps, bounce categorization
- Campaign management with drag-and-drop template builder
- Power BI integration for executive insights
- Event processing pipeline with aggregation services

5. AI Template Generator (Llama 3.3 POC)

- Prompt-to-JSON conversion for Dell Studio Template Builder
- NLP-driven design intent extraction
- Validation & sanitization layer for production readiness

6. DevOps Excellence

- Docker multi-stage builds • PKS/Kubernetes orchestration

- **33% reduction in release cycles** through automated CI/CD
 - Blue-green deployments with zero downtime
 - 80%+ test coverage enforcement
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Full-Time: Software Inventory Agent | 100K+ Device Asset Management

Architecture & Technology Stack

Client: Python, PowerShell • **Server:** REST APIs, RabbitMQ • **Database:** PostgreSQL + TimescaleDB, MongoDB • **Integration:** Flexera, Power BI • **Deployment:** OTA updates, distributed architecture

Core Technical Contributions

1. Intelligent Scanning Engine

Registry Discovery (PowerShell):

HKLM:\SOFTWARE\...\Uninstall* (64-bit + 32-bit)

HKCU:\SOFTWARE\...\Uninstall* (per-user apps)

Get-AppxPackage → UWP/Store apps + browser extensions + PWAs

File System Analysis (Python):

Deep Scan → C:\Users*\{Downloads, Documents, Desktop}

Limited Scan → C:\, Program Files, ProgramData, AppData

Binary signature detection → Portable apps + executables

Metadata extraction → Version, publisher, dependencies

2. Real-Time Telemetry System

Monitoring Capabilities:

- Process execution with parent-child lineage mapping
- Resource tracking: CPU, memory, disk I/O, network latency
- Event detection: installs, uninstalls, crashes, auth failures
- Buffered uploads with 60% compression efficiency
- Offline resilience with local queuing

Telemetry Performance:

Before: Polling-based monitoring → 5-minute granularity, 40% overhead

After: Event-driven telemetry → Real-time (<10s latency), 3% overhead

Achieved through: Windows Event Tracing (ETW) hooks,

Asynchronous I/O (asyncio in Python),
Local SQLite buffer (max 10MB, auto-flush),
zlib compression (level 6, 62% avg compression ratio)

Data Collection Efficiency:

Before: Full snapshots → 2.4 MB/day/device, high bandwidth usage

After: Delta tracking → 0.9 MB/day/device (-62%), bandwidth optimized

Achieved through: Incremental change detection,
binary diff for file modifications,
deduplication (hash-based: SHA-256),
batch uploads (every 15 min, max 100 events)

Offline Resilience:

Buffer capacity: 10,000 events (approx. 3 days of data)

Sync reliability: 99.6% (auto-retry on network restoration)

Data loss rate: <0.1% (persistent local storage)

3. AI-Ready Normalization Engine

Achievements: 75% normalization | 16.6K unique names | 8 GB processed

Techniques:

- └ Fuzzy matching for version standardization
- └ Publisher canonicalization (NLP-based)
- └ Binary signature identification
- └ CVE vulnerability mapping
- └ Machine learning readiness for future automation

Software Normalization Accuracy:

Before: Manual mapping → 42% accuracy, 6 hours/batch processing

After: ML-enhanced fuzzy matching → 75% accuracy (+33 pts), 15 min/batch

Achieved through: Levenshtein distance (threshold=0.85),
TF-IDF similarity for publisher names,
binary hash lookup (SHA-256),
rule-based version parsing (regex + NLP)

Processing Efficiency:

Before: Sequential processing → 2,400 evidences/hour

After: Parallel processing (Python multiprocessing) → 16,000 evidences/hour (+567%)

Achieved through: Worker pool (8 processes), batch size optimization (500),
in-memory caching (LRU cache, 10K entries)

Vulnerability Detection:

CVE matching accuracy: 91% (via NVD integration)

False positive rate: 6% → 2.1% (improved signature validation)

Detection latency: 24 hours → 4 hours (real-time CVE feed)

4. Hybrid Database Architecture

PostgreSQL + TimescaleDB:

```
CREATE TABLE telemetry_events (  
  event_time TIMESTAMPTZ NOT NULL,  
  device_id UUID,  
  event_data JSONB  
);  
SELECT create_hypertable('telemetry_events', 'event_time');
```

Features: JSONB flexibility • Hypertable optimization • Horizontal scaling • Zero licensing cost

MongoDB: Unstructured logs • Fast BSON queries • Schema evolution

5. Microservices & APIs

POST /api/v1/inventory/upload → Bulk inventory submission

POST /api/v1/telemetry/events → Real-time event streaming

GET /api/v1/analytics/utilization → Usage analytics

RabbitMQ Pipeline:

Flexera Integration → Device Scheduling → Agent Execution

→ Verification → Power BI Dashboard

Pilot Success: 130 devices (AMER/EMEA/APJ) • 4K evidences • 2-3 min scans • Daily 10 AM syncs • Seamless OTA updates

Full-Time: Patent - Agentic AI-Driven Software Governance System | USPTO Filing

Patent Overview

Title: Agentic AI-Driven Persona-Based Software Removal and Governance System

Status: Authorized by Dell Technologies for filing with USPTO

Innovation Area: Autonomous endpoint software management with AI-driven decision-making

Patent Abstract

Invented a novel agentic AI system for autonomous software removal and governance that dynamically adapts to user personas, risk contexts, and behavioral patterns. The system addresses limitations in conventional software management tools that rely on static policies, manual configurations, and limited personalization through advanced AI/ML techniques, federated learning, and explainable AI modules.

Core Innovation Architecture

System Components:

1. Data Collection & Aggregation Layer
 - └ Software, hardware, behavioral, and threat intelligence aggregation
2. Agentic AI Decision Engine (Narrow AI Modules)
 - └ Risk Prediction Module (XGBoost, Random Forest, Isolation Forest)
 - └ Adaptive Scheduler (Reinforcement Learning)
 - └ Hypothesis Generation & Testing System (HGTS)
 - └ Context-Aware AI Database (CAAD)
 - └ Persona Profiler AI (K-Means, Clustering)
 - └ AI Decision Controller (Orchestration)
3. Persona-Based Action Layer
 - └ Custom Policy Generator (persona-aligned rules)
 - └ Hyper-Personalization Module (individual exceptions)
 - └ Application Offloading System (cloud/edge migration)
 - └ Approval Workflow Engine (automated approval chains)
4. Execution & Enforcement Layer
 - └ Removal Agent (PowerShell, MSI/EXE/UWP/PWA uninstall)
 - └ Registry & File Cleanup (AI-based residual detection)
 - └ Offload Agent (cloud/secondary storage)
5. Logging, Reporting & Compliance Layer
 - └ Explainable AI Report Generator
 - └ Compliance & Audit Reports
 - └ Real-Time Dashboards & Alerts
6. Federated Learning & Threat Intelligence Sharing
 - └ Federated Model Aggregation
 - └ Community Intelligence Sharing
 - └ Decentralized Threat Intelligence

Key Technical Contributions

1. Agentic AI Decision Engine with Specialized Narrow AI Modules

class AgenticAIEngine:

```
def __init__(self):
    self.risk_predictor = RiskPredictorAI()
    self.scheduler = AdaptiveSchedulerAI()
    self.hypothesis_tester = HypothesisTestingAI()
    self.persona_profiler = PersonaProfilerAI()
    self.offload_decider = OffloadDecisionAI()
    self.artifact_detector = ResidualArtifactDetectorAI()
    self.decision_controller = AIDecisionController()

def evaluate_endpoint(self, endpoint_data, user_profile):
    risk_scores = self.risk_predictor.predict(endpoint_data)
    persona_info = self.persona_profiler.profile(user_profile)
    scan_schedule = self.scheduler.adjust(persona_info, system_load)
    anomalies = self.hypothesis_tester.detect(behavioral_logs)
    offload_recs = self.offload_decider.recommend(usage_stats)

    actions = self.decision_controller.synthesize(
        risk_scores, anomalies, persona_info, offload_recs
    )
    return actions
```

AI/ML Models Employed:

- **Supervised Learning:** Random Forests, XGBoost for software classification (authorized/unauthorized/redundant)
- **Unsupervised Learning:** K-Means Clustering, Isolation Forests for outlier detection
- **Anomaly Detection:** Autoencoders, One-Class SVM for residual artifact identification
- **Reinforcement Learning:** Policy optimization for adaptive scheduling via reward-feedback loops
- **Federated Learning:** Federated Averaging for decentralized model updates

2. Persona-Based Intelligence System

User Persona Mapping:

- └ Developer Persona: Allow IDEs, compilers, Docker → block gaming apps
- └ Executive Persona: Allow productivity suite → block dev tools
- └ Analyst Persona: Allow BI tools, Python → block design software
- └ Custom Exceptions: Individual usage patterns override persona defaults

Hyper-Personalization:

- Behavioral analytics from clickstream data

- Application usage duration tracking
- Idle time pattern analysis
- Access control metadata integration

3. Advanced Residual Cleanup Mechanism

Registry Cleanup Coverage:

HKEY_LOCAL_MACHINE\SOFTWARE*
HKEY_CURRENT_USER\SOFTWARE*
HKEY_CLASSES_ROOT*
HKEY_USERS*

AI-Based Artifact Detection:

- Anomaly models trained on registry patterns
- Hidden file discovery via filesystem traversal
- Orphaned process identification
- Leftover service detection

4. Federated Learning Framework

Privacy-Preserving Intelligence:

- └ Local model training on anonymized endpoint data
- └ Federated Aggregator combines distributed updates
- └ Threat Sharing API for software fingerprints
- └ Global CAAD Model updates without raw data transfer
- └ Differential privacy mechanisms (ϵ -differential privacy)

Performance Metrics & Impact

AI Model Performance:

Risk Prediction Accuracy:

Before: Static blocklists → 61% detection rate, 34% false positives

After: XGBoost ensemble → 87% detection (+26 pts), 8% false positives (-26 pts)

Achieved through: Supervised learning on 50K+ labeled software instances,
behavioral features (30+ dimensions), threat intel integration,
ensemble voting (3 models), continuous retraining

Persona Classification Accuracy:

Before: Manual role assignment → 72% accuracy, quarterly updates

After: K-Means clustering + behavioral analysis → 91% accuracy (+19 pts)

Achieved through: Clickstream analysis, app usage patterns,

access control metadata, hierarchical clustering,
silhouette score optimization (0.68 avg)

Anomaly Detection for Unauthorized Software:

Before: Signature-based detection → 58% catch rate, high false negatives

After: Isolation Forest + Autoencoder → 84% catch rate (+26 pts), 7% false positives

Achieved through: Unsupervised learning on 100K+ endpoint snapshots,
feature engineering (process behavior, network activity),
ensemble anomaly scoring, adaptive thresholds

Residual Artifact Detection:

Before: Static cleanup scripts → 45% leftover artifacts remaining

After: AI-powered deep cleanup → 94% cleanup rate (+49 pts)

Achieved through: Registry pattern learning (neural networks),
filesystem anomaly detection, orphaned process identification,
trained on 20K+ uninstallation events

System Efficiency Improvements:

Software Removal Efficiency:

Before: Manual intervention → 8 hours/100 devices, 68% success rate

After: Autonomous agentic system → 25 minutes/100 devices (-97%), 96% success

Achieved through: Parallel execution, adaptive scheduling,
intelligent retry logic, context-aware timing

Compliance & Audit:

Before: Quarterly manual audits → 15% coverage, 40 hours/audit

After: Real-time compliance tracking → 100% coverage, 2 hours/report

Achieved through: Automated logging, explainable AI reports,
continuous monitoring, persona-specific justifications

Resource Optimization:

Storage reclaimed: Average 8.2 GB/device (unauthorized software removal)

License cost savings: \$2.3M annually (unused software identification)

Security incident reduction: 42% decrease in malware from unauthorized apps

Admin time saved: 75% reduction in manual software management tasks

Federated Learning Performance:

Threat Intelligence Sharing:

Model convergence: 85% accuracy after 5 federated rounds (vs. 78% centralized)

Privacy preservation: Zero raw data transfer, $\epsilon=0.1$ differential privacy

Global threat detection: 23% improvement in zero-day software detection

Cross-organization learning: 12 participating organizations, 50K+ endpoints

Adaptive Scheduling Intelligence:

Scan Frequency Optimization:

Before: Fixed schedules → 40% scans during active usage, user disruption

After: RL-based adaptive scheduling → 95% scans during idle periods

Achieved through: Reinforcement learning with user activity rewards,
system load monitoring, historical success rate analysis,
Q-learning for optimal timing policy

CPU Impact During Scans:

Before: 28% average CPU usage during scans

After: 6% average CPU usage (-79%) via intelligent resource throttling

Patent Claims & Novel Contributions

Key Patent Claims (10 Total):

1. **Independent Claim:** System for autonomous software removal with Agentic AI engine, persona profiling, and federated learning
2. **ML Model Claim:** Random Forest, XGBoost, K-Means, Isolation Forest, Autoencoder implementations
3. **Persona Logic Claim:** Dynamic persona assignment based on usage frequency, idle time, interaction patterns
4. **Adaptive Scheduling Claim:** Context-aware task execution based on CPU load, user availability, historical success
5. **Multi-Format Removal Claim:** MSI, EXE, portable apps, UWP, PWA uninstallation capabilities
6. **Artifact Detection Claim:** Anomaly-based residual software artifact identification
7. **Approval Workflow Claim:** Automated user/admin confirmation workflows
8. **Offloading Mechanism Claim:** Cloud/secondary storage migration based on persona thresholds
9. **Explainability Claim:** Human-readable AI decision justifications per persona
10. **Federated Intelligence Claim:** Distributed model updates via federated averaging

Strategic Business Impact

Dell Technologies Strategic Alignment:

- **Autonomous Device Hygiene:** Zero-touch endpoint management for Dell Client Command Suite

- **AI-Augmented Security:** Strengthens Dell's endpoint security portfolio with behavioral anomaly detection
- **Persona-Centric UX:** Enhances Dell Optimizer and Unified Workspace offerings
- **Market Differentiation:** Novel IP strengthening Dell's defensible innovation portfolio
- **Cross-Solution Integration:** Seamless integration with Dell Data Security, Workspace ONE

Competitive Advantages:

- First-to-market with agentic AI for software governance
- Federated learning approach preserves privacy while sharing threat intelligence
- Explainable AI ensures regulatory compliance (GDPR, SOX, HIPAA)
- Hyper-personalization reduces user disruption vs. rule-based competitors

Intellectual Property Value:

- USPTO filing authorized (provisional/non-provisional application)
- Defensive patent portfolio against competitors (Microsoft Intune, VMware Workspace ONE)
- Potential licensing opportunities for enterprise software management vendors
- Strengthens Dell's AI/ML patent portfolio in endpoint management domain

Technical Innovation Summary

Novel Contributions to State-of-the-Art:

1. **Agentic Autonomy:** Self-directed AI agents vs. rule-based automation
2. **Persona-Based Governance:** Dynamic policy adaptation per user role/behavior
3. **Federated Threat Intelligence:** Privacy-preserving collaborative learning
4. **Explainable Remediation:** AI decision transparency for compliance
5. **Hypothesis-Driven Detection:** Proactive anomaly identification vs. reactive scanning
6. **Reinforcement Learning Scheduler:** Adaptive timing optimization via reward signals

Problem Statements Addressed:

- Static policies lacking personalization → Dynamic persona-driven policies
- Outdated software identification → AI-driven continuous learning
- Fixed scheduling → Adaptive scheduling based on context
- Shallow cleanup routines → AI-powered deep residual detection
- No audit trails → Comprehensive explainable AI reporting
- Isolated threat intelligence → Federated collaborative learning

Technical Proficiency Matrix

Domain	Technologies
AI/ML	Azure OpenAI (GPT-4/5), Llama 3.3, XGBoost, TensorFlow, PyTorch, Scikit-Learn, Sentence-BERT, BERTopic, spaCy, LangChain, MLflow, Random Forest, Isolation Forest, Autoencoders, One-Class SVM, K-Means, Reinforcement Learning (Q-Learning), Federated Learning
Languages	Python, Go, JavaScript/TypeScript, PowerShell
Frontend	React.js (hooks, ATOM architecture, state management, drag-and-drop UI)
Backend	Node.js (Express), Go (goroutines, REST APIs), FastAPI, Flask, GraphQL
Cloud & Data	Azure (Data Factory, Synapse, ML, OpenAI, Data Lake Gen2, DevOps), Delta Tables, Feature Store, Terraform
Databases	PostgreSQL, TimescaleDB, MongoDB, Redis, Greenplum, SQL Server
Search & Cache	Elasticsearch, Redis
Infrastructure	RabbitMQ, Docker, Kubernetes (AKS, PKS), CI/CD (GitLab, Azure DevOps), SMTP
Analytics	Power BI, Tableau, statistical modeling, causal inference
Security & Compliance	EDR/XDR integration, threat intelligence, differential privacy, explainable AI, audit logging
Methodologies	Agile, Microservices, Full-Stack Development, Atomic Design (ATOM), Federated Learning, Agentic AI Systems

Career Progression & Impact

Internship Journey

Summer Internship → User360 Dashboard with 40% deployment optimization ↓ **Winter Internship** → AI-enabled security & training portals with 90%+ CI/CD maturity ↓ **FTE Offer** → Recognition of technical excellence and project impact

Full-Time Achievements

Category	Achievement
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Cost Savings	\$25M (Communique) + \$2M (license optimization) = \$27M annually
ML Accuracy	Engagement prediction: 52% → 78% • Normalization: 42% → 75% • Search relevance: 67% → 89%
Performance	API latency: 3.2s → 0.4s (-88%) • Go service: 850ms → 180ms (-79%) • Upload throughput: +611%
Efficiency	33% faster releases • 90% audit reduction • 60% template creation speedup • 40% deployment time reduction
Scale	100K+ devices • 16.6K software IDs • 10K emails/min sustained • 340 API req/sec
Quality	90%+ CI/CD maturity • 82% code coverage • 99.9% delivery • Zero-downtime deploys
AI/ML Impact	Open rate: +39% • CTR: +78% • Recommendation CTR: +78% • Search queries reduced: 8 → 2.3

Key Innovations

AI/ML Leadership

- Multi-armed bandit for dynamic personalization
- Cox regression for attrition prediction
- LLM-powered KPI assistant with natural language generation
- Hybrid recommendation engine (collaborative + content-based)
- Reinforcement learning feedback loops
- Semantic search with vector embeddings
- MLflow-based model versioning and A/B testing

System Architecture

- Go-powered SharePoint migration for cost optimization
- TimescaleDB time-series optimization for telemetry
- Fault-tolerant distributed agent architecture
- GraphQL for optimized data querying
- Priority queue architecture for critical communications
- Real-time telemetry with offline buffering
- ATOM architecture for scalable frontend components

DevOps & Infrastructure

- GitLab and Azure DevOps CI/CD pipelines
 - Kubernetes (AKS/PKS) orchestration
 - Infrastructure as Code (Terraform)
 - Blue-green deployments
 - Automated testing gates and security scanning
-

Professional Skills & Competencies

Technical Excellence

- **Full-Stack Development:** End-to-end ownership from database to UI
- **AI/ML Engineering:** Production ML pipelines with real-world impact
- **Cloud Architecture:** Azure-native solutions with cost optimization
- **DevOps Practices:** CI/CD maturity transformation (90%+ scores)
- **Database Design:** Hybrid architectures for optimal performance
- **API Development:** REST, GraphQL, and high-performance Go services

Soft Skills

- **Leadership:** Led development of critical features and AI initiatives
 - **Collaboration:** Cross-functional work with security, HR data, and analytics teams
 - **Problem Solving:** Transformed failing pipelines, optimized deployments, delivered \$27M value
 - **Agile Methodology:** Sprint planning, iterative development, continuous improvement
 - **Communication:** Technical documentation, stakeholder presentations, code reviews
-

Portfolio demonstrates comprehensive software engineering expertise from internships through full-time employment, with proven track record of delivering enterprise-scale AI/ML solutions, distributed systems, and measurable business value.

Dell Software Management Agent | Real-Time Remote Software Control

Problem Statement

Dell's IT infrastructure lacked centralized, real-time software management across 100K+ devices. Manual interventions required 24-48 hours for software removal, creating security risks, compliance gaps, and 15+ hours/week IT overhead.

Business Impact: *Security vulnerabilities, compliance violations, operational inefficiency across global enterprise infrastructure.*

Architecture & Technology Stack

Client: *.NET 8.0 Native AOT (C#) • Windows Services • SignalR • PowerShell*

Server: *ASP.NET Core 8.0 • SignalR Hub • PCF • WebSockets/SSE*

Deployment: *Self-contained executables • Silent Windows Service • OTA updates*

Core Technical Contributions

1. Real-Time Pub/Sub System

SignalR Architecture: *Server (PCF) → WebSocket/SSE → 100K+ Client Agents → Execute → ACKs/Results*

Performance:

- **Latency:** *<3s command to execution |* **Success Rate:** *99.7%*
- **Scalability:** *720+ concurrent connections/instance (1GB memory)*
- **Auto-reconnect:** *99.4% success with exponential backoff*

Impact:

Before: Batch processing → 24-48 hour cycles, 500 devices/batch

After: Real-time execution → <5s avg response, unlimited concurrent

Achieved: WebSocket persistent connections, multi-transport fallback,

connection pooling, memory optimization

2. Native AOT Agent

Specifications: 25-30 MB executable • 50-70 MB memory • <1s startup • <0.5% CPU idle

Efficiency:

Package Size | Install Time | Memory | Dependencies

Before (Runtime): 350 MB | 5-10 min | 150-200 MB | .NET Runtime

After (Native AOT): 30 MB | <30 sec | 60 MB | None

Improvement: -91% | -83% | -67% | Zero conflicts

Production Impact: 95% bandwidth savings, 10× faster deployment, zero runtime updates

3. Intelligent Command Dispatcher

Multi-Mode Execution:

- **EXECUTE_AGENT** → Dell Software Removal Agent
- **LAUNCH_PROGRAM** → Any executable with arguments
- **CUSTOM** → PowerShell scripts, batch files

Capabilities: MSI/non-MSI apps • UWP/Store apps • Portable apps • Registry cleanup

Metrics:

- Execution: 2-4s avg | Success: 94.3% | Auto-retry on failure
- Resource: BelowNormal priority, GC every 30 min, <3% CPU during execution

4. Advanced Targeting & Dashboard

Targeting: Broadcast (All) • Group (by hostname) • Individual (specific device)

Real-Time Monitoring:

- Live client status (connection health, uptime)
- Command execution tracking (per-device progress)
- Activity logs with filtering (success/failure/warnings)
- Performance metrics (memory, CPU, network)

SLA: 99.9% uptime • 99.7% command delivery • <5s P95 latency

5. Security & Compliance

Security: HTTPS/WSS encryption • LocalSystem service • Full audit trail

Compliance: Complete logging (timestamp, user, target, result) • 90-day retention • Change management history

Production Metrics & Impact

Deployment Scale

100,000+ devices (AMER/EMEA/APJ) • **5,000+ commands/day** • **720+ concurrent connections/instance**

Operational Efficiency

Metric	Before	After	Improvement
Software Removal Time	24-48 hrs	<5 sec	99.9%
IT Labor/Week	15+ hrs	<2 hrs	87%
Deployment Bandwidth	350 MB	30 MB	91%
Installation Time	5-10 min	<30 sec	95%
Manual Interventions	100%	<3%	97%

Cost Savings

IT Labor: 13 hrs/week saved × 52 weeks × \$75/hr = \$50,700/year

Bandwidth: 100K devices × 320 MB × \$0.05/GB = \$16,000/year

Total: ~\$67,000/year

ROI: 12 months

System Performance

Component	Metric	Value
Server	Memory	512MB-1GB/instance
	CPU	<5% idle, 15-25% load
	Connections	720+/instance
Client	Memory	50-70 MB
	CPU	<0.5% idle, 2-3% exec
	Startup	<1 second

Reliability

99.9% Uptime • 94.3% Command Success • 99.4% Client Connectivity • <0.1% Data Loss

Business Impact

Security: Real-time threat response • Instant compliance enforcement • Audit-ready logging

Operations: 87% labor reduction • 99.9% faster deployments • Zero-touch automation • Global reach (<5s latency)

Cost: \$67K/year savings • 91% bandwidth reduction • Zero runtime licensing
Scale: 100K+ devices • Linear scaling • PCF auto-scaling • High availability

Technical Innovations

1. **SignalR at Scale:** Real-time bidirectional communication for 100K+ devices with <5s latency
 2. **Native AOT:** 91% size reduction, 67% memory savings, zero dependencies
 3. **Hybrid Execution:** MSI, non-MSI, UWP, portable apps, scripts in single agent
 4. **Production Dashboard:** Live monitoring with real-time status and audit trails
 5. **Enterprise Deployment:** PCF with auto-scaling, health checks, zero-downtime updates
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Status:  **Production** | 100,000+ devices | 99.9% uptime | 5,000+ daily commands

Timeline: 4 months (Architecture → Development → Production Rollout)

Team: 1 Software Engineer (Full-Stack + DevOps)

Demonstrates expertise in distributed systems, real-time communication, enterprise deployment, performance optimization, and production-grade engineering.